Team Number: 25

VANGUARD: COMMUNITY STEWARDSHIP

M2 - Mid-Submission

1. Current Approach and Changes Since M1

RupeeRise continues with the same core goal stated in M1: a student-centered financial-literacy web application where learners practise safe money decisions through fictional rupees, relatable Hyderabad personas, simulated consequences, quizzes and supportive feedback. It remains strictly educational: no real bank accounts, no real investment recommendations and no promised returns.

The main change since M1 is the implementation stack. M1 proposed a React frontend, Node.js/Express backend and Supabase/PostgreSQL. The GitHub repository now implements the product as a React 19 and Vite frontend with a FastAPI modular-monolith backend, PostgreSQL, SQLAlchemy and Alembic migrations. This change is appropriate because the project now depends heavily on deterministic financial engines, auditable scoring rules, server-side validation and a broader automated test suite.

The current approach makes the server authoritative. The browser presents the journey, while the backend validates transitions, calculates outcomes, records progress, protects assessment integrity and exposes educator views. Learning progress, badges, reward wallet entries and fictional portfolio activity are kept separate so students are rewarded for understanding rather than luck.

Implemented learning areas now include Plan and Budget, Save and Manage, Understand Investing, Stay Financially Safe, the Rs.500 Money Lab, Banking Explorer, reflections, a fictional reward wallet, Portfolio Quest and a public journey demo. The system also supports grade-derived stages, namely Explorer, Builder, Future Planner and Investor Sandbox, with the backend enforcing stage eligibility. The interface includes English, Hindi and Telugu UI chrome, though learning content remains English-only at this stage.

Code push: the local checkout was fast-forwarded to GitHub main at commit **4d3fc77**, titled *Security hardening: invites, cookie sessions, isolation, private answer keys, tutor safety, deployment*. Therefore this M2 draft reflects the current GitHub state.

2. Experiments Run So Far and Their Results

- **Deterministic simulation engine:** budget, savings, investing, safety and Money Lab experiences use server-side rules. Reproducible seeds allow attempts to be replayed, and investing scores are based on reasoning rather than market luck.
- **Learning and assessment model:** the repository contains one starting check, one final check and four module quizzes. Misconception tagging, rapid-answer dampening, retake weighting and mastery tracking prevent progress from being only a completion count.
- **Student engagement experiments:** achievements, the fictional reward wallet, Portfolio Quest, reflections, Money Lab and Banking Explorer make practice active while keeping learning scores separate from fictional money.
- **Security hardening:** educator registration now uses invitation tokens instead of a shared code; sessions use HttpOnly cookies with CSRF protection; refresh rotation, session revocation, password-change handling, organisation isolation and rate limits were added.
- **Content privacy:** public content JSON no longer carries answer keys. A private answer-key bundle is merged at startup, while development/test mode uses labelled placeholders.

3. Progress on Testing

The repository's latest testing notes report 417 backend tests passing with the private content bundle, or 410 passing and 7 skipped without it. The same notes report 31 frontend unit/component tests and 22 Playwright end-to-end journeys with the private bundle. Coverage includes onboarding, dashboard navigation, the four pillars, quiz flow, educator cohort creation, student join flow and an automated accessibility scan.

During this M2 drafting pass, a lightweight verification was also run against the current checkout. Repository hygiene passed, backend ruff checks passed, and the frontend TypeScript check passed after refreshing the newly added anime.js dependency. The full backend, frontend and Playwright matrix was not rerun here, so the larger numbers above are latest recorded repository results, not new evidence from this PDF step.

Known gaps are being carried forward honestly. No 1,500-student load test has been run yet; Docker/nginx were not verified here; no manual screen-reader, keyboard-only or cross-browser review is complete; the live AI tutor path is unverified without an API key; and learning gains have not yet been measured with real students.

4. Plan for the Remaining Time

- **Final verification:** rerun the full test matrix with and without the private content bundle: backend tests, Alembic checks, frontend tests, production build, Playwright journeys and axe accessibility checks.
- **Deployment rehearsal:** verify the chosen hosting setup with managed PostgreSQL, production checks, HTTPS, cookie security, private-content mounting and seeded demo data.
- **Load and accessibility:** perform a small load test toward the 1,500-student target and run one manual keyboard/screen-reader pass on the student journey and educator dashboard.
- **Content and demo polish:** prepare a repeatable demo path from onboarding to one simulation, reflection, reward, wallet/portfolio view and educator dashboard, while stating limitations plainly.
- **Submission hygiene:** keep the private answer-key bundle out of GitHub, add replacement question variants for exposed historical keys, and attach deliverables/M2.pdf to the repository.

Overall, M2 shows that the project has moved from a planned concept into a working full-stack prototype. The remaining risk is not the absence of features; it is proving the prototype under deployment, accessibility, load and real-student conditions before final submission.